Meera Iyer

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Professional Summary

NLP Engineer with two years of experience fine-tuning transformer models for Indian-language text classification and entity extraction. Strong in dataset annotation, evaluation and error analysis; serves models through FastAPI and Docker. Limited experience with large language model applications beyond prompt experiments.

Technical Skills

Programming: Python, SQL

NLP: Hugging Face Transformers, BERT and XLM-RoBERTa fine-tuning, spaCy, named entity recognition, text classification, tokenisation for Tamil and Hindi

Machine Learning: PyTorch, scikit-learn, evaluation (precision, recall, F1, confusion analysis), active learning

Data: Annotation guidelines, Label Studio, inter-annotator agreement

Deployment: FastAPI, Docker, MLflow experiment tracking, ONNX export (basic)

LLM Exposure: Prompt experiments with hosted LLM APIs for data labelling assistance

Work Experience

NLP Engineer - Tamarind Language Technologies Pvt Ltd
Chennai, Tamil Nadu | Jul 2024 - Present

- Fine-tuned XLM-RoBERTa for complaint classification in Tamil, Hindi and English across 14 categories; macro F1 rose from 0.71 (TF-IDF baseline) to 0.86.
- Developed a named entity recognition model for addresses and product names in code-mixed text using a spaCy transformer pipeline; entity F1 reached 0.91 on the test set.
- Created annotation guidelines and ran Label Studio sessions with four annotators; improved Cohen's kappa from 0.68 to 0.82 after guideline revision.
- Packaged models as FastAPI services in Docker and tracked experiments with MLflow; p95 latency 120 ms on CPU after quantisation.
- Used LLM prompts to pre-label 5,000 samples, which human annotators then corrected, saving about 35% of labelling time.

Projects

Code-mixed Sentiment Analysis

- Problem: Customer reviews mix English with Tamil and Hindi in Roman script.
- Approach: Normalised transliteration variants, fine-tuned a multilingual transformer and examined failures by language mix.
- Technologies: PyTorch, Hugging Face, Python
- Results: Weighted F1 0.84; error analysis identified sarcasm and negation as top failure types.

Resume Skill Extraction (Academic)

- Problem: Extract skill phrases from semi-structured resumes.
- Approach: Annotated 400 documents, trained a spaCy NER model and added rule-based post-processing for abbreviations.
- Technologies: spaCy, Label Studio, Python
- Results: Entity F1 0.83 on a held-out set of 80 resumes.

Education

M.Tech, Data Science and Engineering, Coromandel Institute of Technology, Chennai (2022 - 2024) - CGPA 8.9 / 10

B.E., Computer Science and Engineering, Marina College of Engineering, Chennai (2018 - 2022) - CGPA 8.2 / 10

Certifications

- Natural Language Processing Specialization (Coursera, 2023)

- Hugging Face NLP Course (completed, 2023)

Achievements

- Paper on code-mixed sentiment accepted at a regional NLP workshop, 2024.
- Reduced labelling effort by 35% with LLM-assisted pre-labelling.